SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 0604 **COURSE NAME:** MACHINE LEARNING

APPLICATION BASED QUESTIONS

1. **In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated.**

   Let

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   -

   Probability of selecting an adulterated bottle:

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   Using Bayes' Theorem,

   Substitute values:

   Therefore, the probability that the adulterated bottle came from Shop Y is approximately 52.63%.

2. **A warehouse has two sections. Section A contains 15 defective items and 35 non-defective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective.**

Given:

Selection probabilities:

Defective probabilities:

Using Bayes' Theorem,

Substitute values:

3. In a binary classification problem using a deep learning model with a K-Nearest Neighbor classifier, the model produces True Positive (TP) = 120 and True Negative (TN) = 150.The dataset contains 1000 samples per class, and 20% of the data is used for testing.

Given Data

- TP = 120
  TN = 150
- Each class has **1000 samples**, and **20%** is used for testing.
  Calculate:
- Accuracy
- Precision
- Sensitivity (Recall)
- Specificity
- **Step 1: Test samples**
- Positive class:
-
- Negative class:
-

- Total testing samples = **400**
- **Step 2: Find FN and FP**
- Positive samples:
- Negative samples:
- **Step 3: Calculate Metrics**
- **Accuracy**
- **Precision**
- **Sensitivity (Recall)**
- **Specificity**
- Accuracy = 67.5%
- Precision = 70.59%
- Sensitivity (Recall) = 60%
- Specificity = 75%